

Workshop on Statistics and Machine Learning in Practice

Topic: Unsupervised learning clustering

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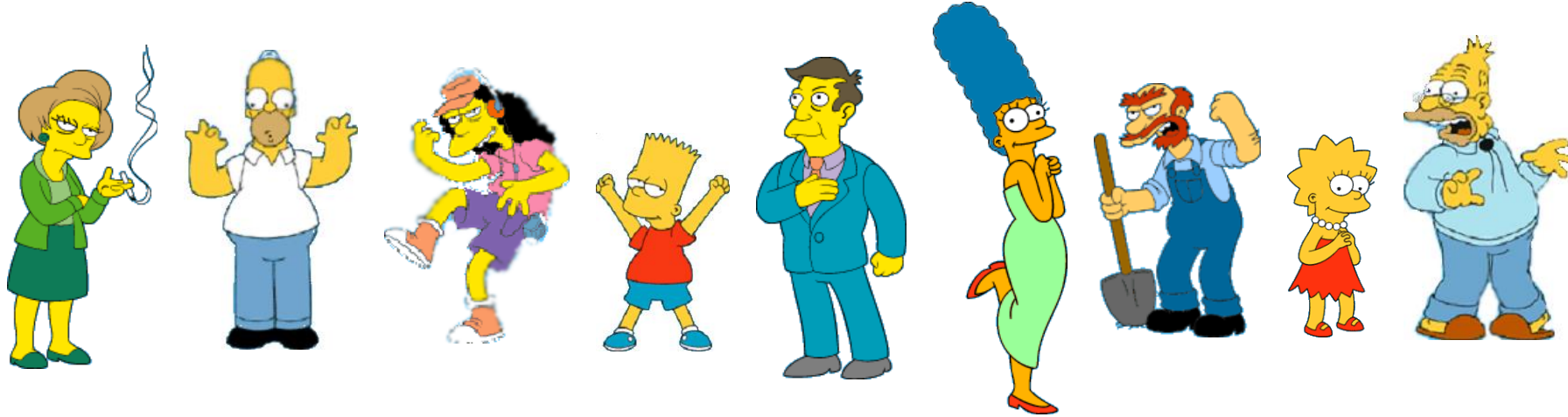
by

Swarup Chattopadhyay, ISI, Kolkata

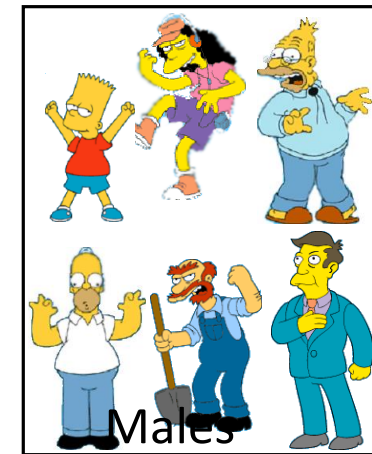
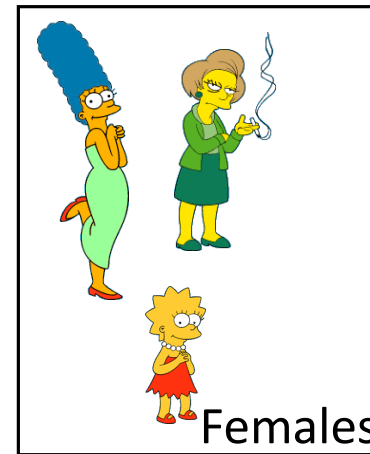
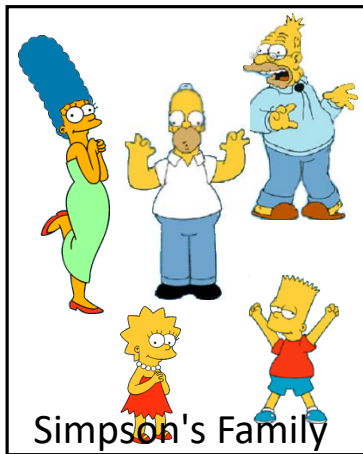
What is clustering?

- The organization of unlabeled data into similarity groups called clusters.
- A cluster is a collection of data items which are “similar” between them, and “dissimilar” to data items in other clusters.
 - high intra-class similarity
 - low inter-class similarity
- More informally, finding natural groupings among objects.

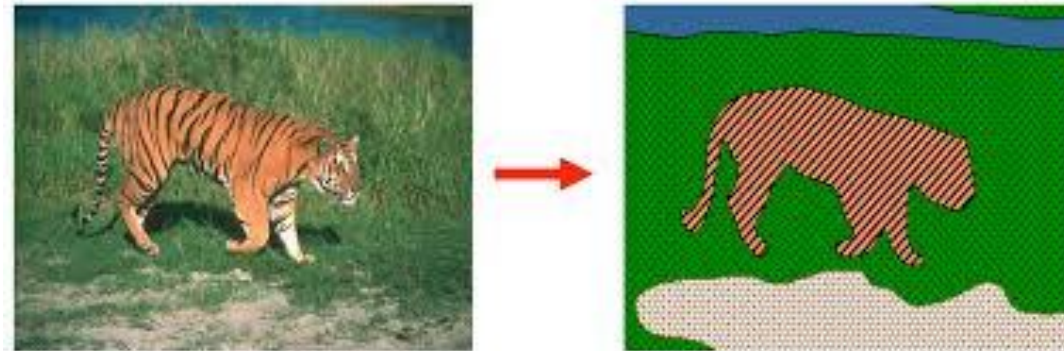
What is a natural grouping among these objects?



Clustering is subjective



Computer vision application : Image segmentation

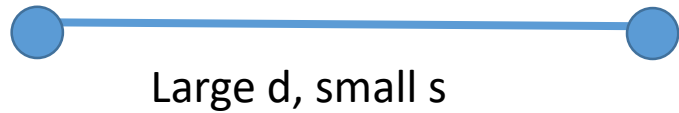


Data Representations for Clustering

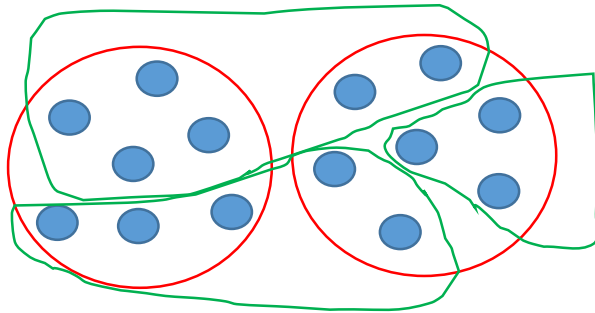
- Input data to algorithm is usually a vector (also called a “tuple” or “record”)
- Types of data
 - Numerical
 - Categorical
 - Boolean
- Example: Clinical Sample Data
 - Age (numerical)
 - Weight (numerical)
 - Gender (categorical)
 - Diseased? (boolean)
- Must also include a method for computing similarity of or distance between vectors

What do we need for clustering?

- Proximity measure, either
 - Similarity measure $s(X_i, X_k)$: large if X_i, X_k are similar
 - Dissimilarity (or distance) measure $d(X_i, X_k)$: small if X_i, X_k are similar



- Criterion function to evaluate clustering



- Algorithm to evaluate clustering
 - For example by optimizing criterion function

Distance (dissimilarity) measures

- Euclidean distance
 - Translation invariant

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^d (x_i^k - x_j^k)^2}$$

- Manhattan (city block) distance
 - Approximation to Euclidean distance
 - Cheaper to compute

$$d(x_i, x_j) = \sum_{k=1}^d |x_i^k - x_j^k|$$

- They are special cases of Minkowski distance:
 - p is a positive integer

$$d_p(x_i, x_j) = \left(\sum_{k=1}^d |x_i^k - x_j^k|^p \right)^{\frac{1}{p}}$$

Cluster evaluation (a hard problem)

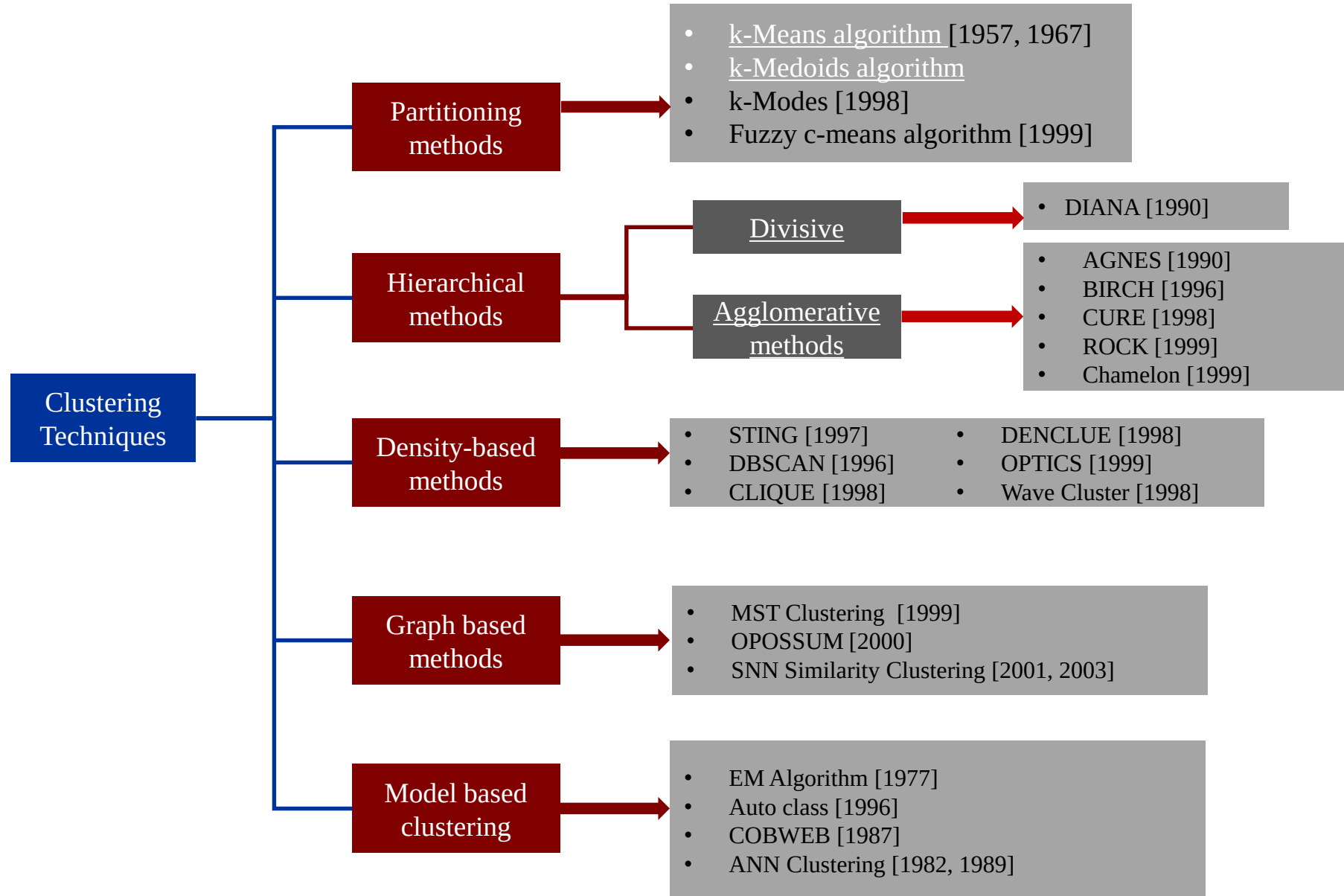
- **Intra-cluster cohesion** (compactness):
 - Cohesion measures how near the data points in a cluster are to the cluster centroid.
 - Sum of squared error (SSE) is a commonly used measure.
- **Inter-cluster separation** (isolation):
 - Separation means that different cluster centroids should be far away from one another.
- In most applications, expert judgments are still the key

How many clusters?



- Possible approaches
 - Fix the number of cluster to k
 - Find the best clustering according to the criterion function

Clustering techniques

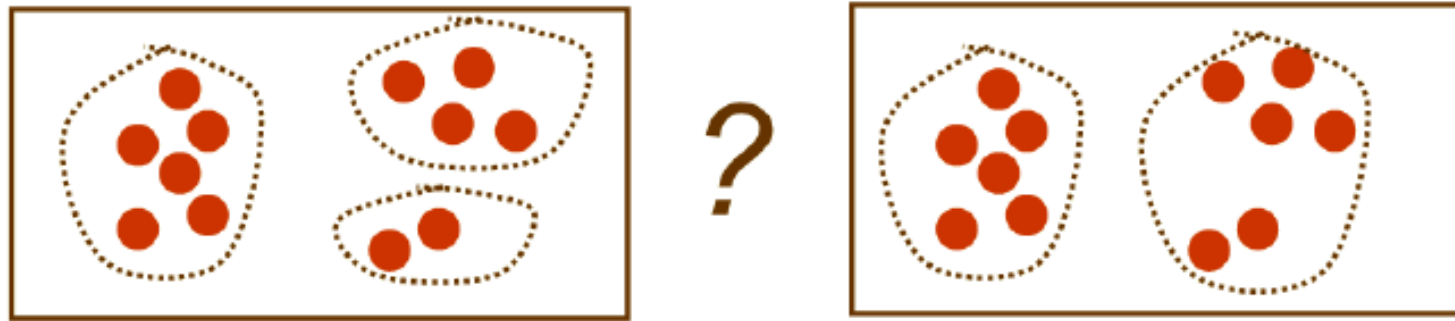


Clustering techniques

- In this lecture, we shall try to cover the following clustering techniques only.
 - Partitioning
 - k-Means algorithm
 - PAM (k-Medoids algorithm)
 - Hierarchical
 - Divisive algorithm
 - Agglomerative algorithm

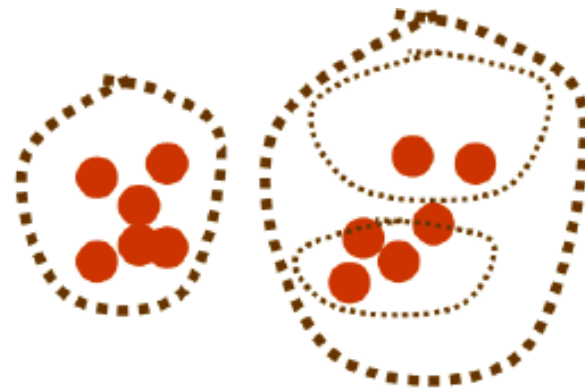
Hierarchical clustering

- Consider flat clustering

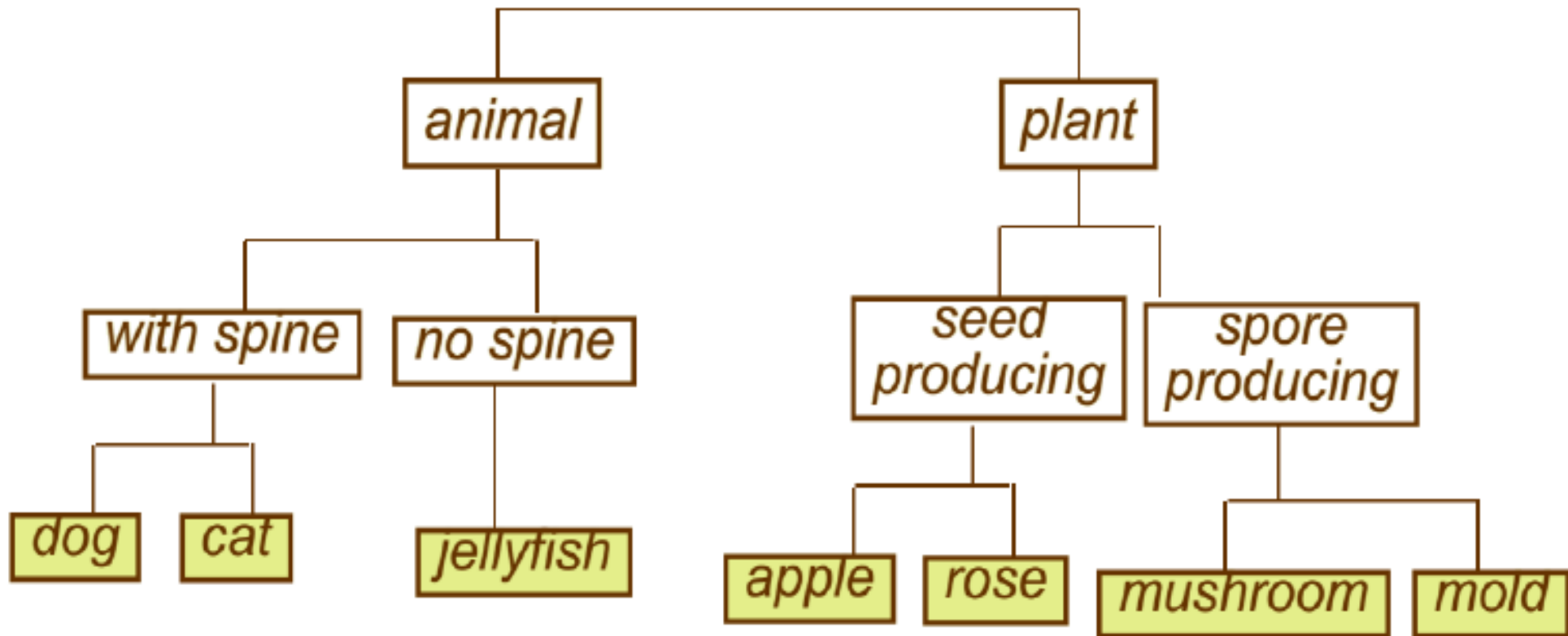


- For some data hierarchical clustering more appropriate than ‘flat’ clustering

- Hierarchical clustering



Example: biological taxonomy



Types of hierarchical clustering

Divisive (top down) clustering

- Starts with all data points in one cluster, the root, then
 - Splits the root into a set of child clusters. Each child cluster is recursively divided further
 - stops when only singleton clusters of individual data points remain, i.e., each cluster with only a single point

Agglomerative (bottom up) clustering

- The dendrogram is built from the bottom level by
 - merging the most similar (or nearest) pair of clusters
 - stopping when all the data points are merged into a single cluster (i.e., the root cluster).
- The number of dendrograms with n leafs = $(2n - 3)! / [(2^{n-2}) (n - 2)!]$

Summary of Hierarchical Clustering Methods

- No need to specify the number of clusters in advance.
- Hierarchical nature maps nicely onto human intuition for some domains
- They do not scale well: time complexity of at least $O(n^2)$, where n is the number of total objects.
- Like any heuristic search algorithms, local optima are a problem.
- Interpretation of results is (very) subjective.

K-Means clustering

- K-means (MacQueen, 1967) is a partition clustering algorithm
- Let the set of data points D be $\{X_1, X_2, \dots, X_n\}$, where $X_i = (X_{i1}, X_{i2}, \dots, X_{ir})$ is a vector in $X \subseteq R^r$, and r is the number of dimensions.
- The k-means algorithm partitions the given data into k clusters:
 - Each cluster has a cluster center, called centroid.
 - k is specified by the user

K-Means clustering

- Given k , the k -means algorithm works as follows:
 1. Choose k (random) data points (**seeds**) to be the initial **centroids**, cluster centers
 2. Assign each data point to the closest **centroid**
 3. Re-compute the **centroids** using the current cluster memberships
 4. If a convergence criterion is not met, repeat steps 2 and 3

K-Means clustering

Input: D is a dataset containing n objects, k is the number of cluster

Output: A set of k clusters

Steps:

1. Randomly choose k objects from D as the initial cluster centroids.
2. For each of the objects in D do
 - Compute distance between the current objects and k cluster centroids
 - Assign the current object to that cluster to which it is closest.
3. Compute the “cluster centers” of each cluster. These become the new cluster centroids.
4. Repeat step 2-3 until the convergence criterion is satisfied
5. Stop

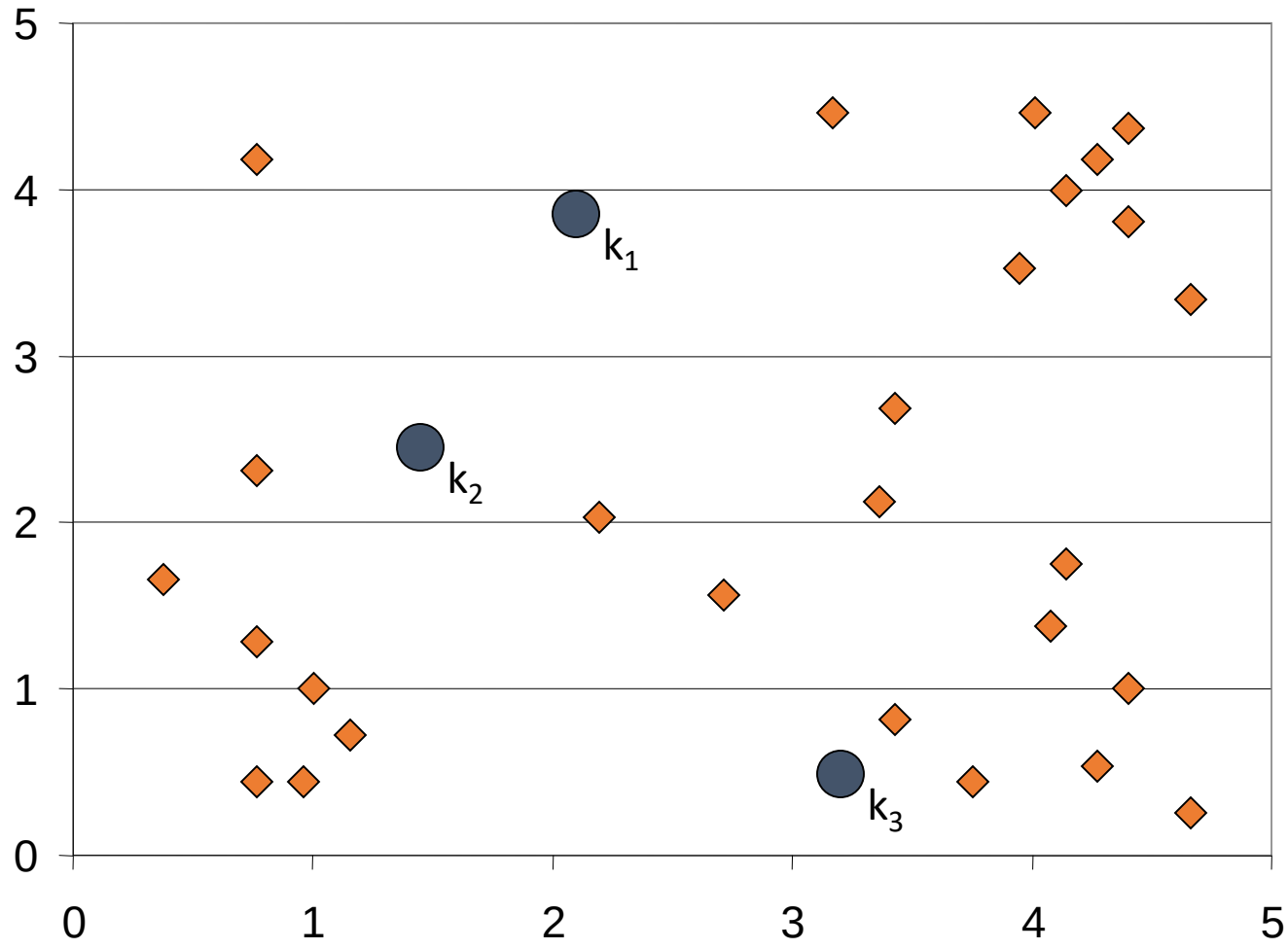
K-means convergence (stopping) criterion

- no (or minimum) re-assignments of data points to different clusters, or
- no (or minimum) change of centroids, or
- minimum decrease in the sum of squared error (SSE),

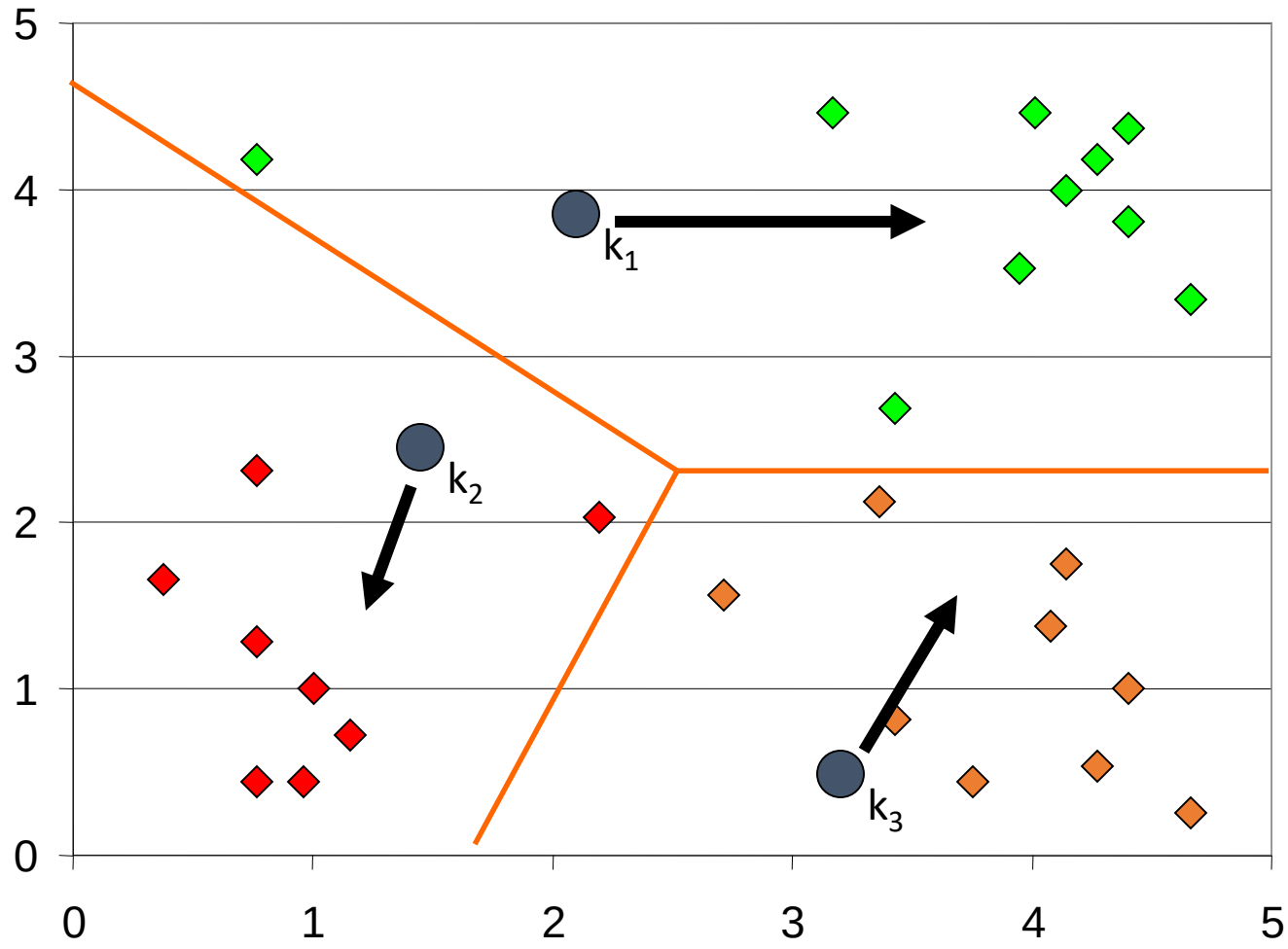
$$SSE = \sum_{j=1}^k \sum_{X \in C_j} d(X, m_j)^2$$

- C_j is the j^{th} cluster,
- m_j is the centroid of cluster C_j (the mean vector of all the data points in C_j),
- $d(X, m_j)$ is the (Euclidean) distance between data point x and centroid m_j .

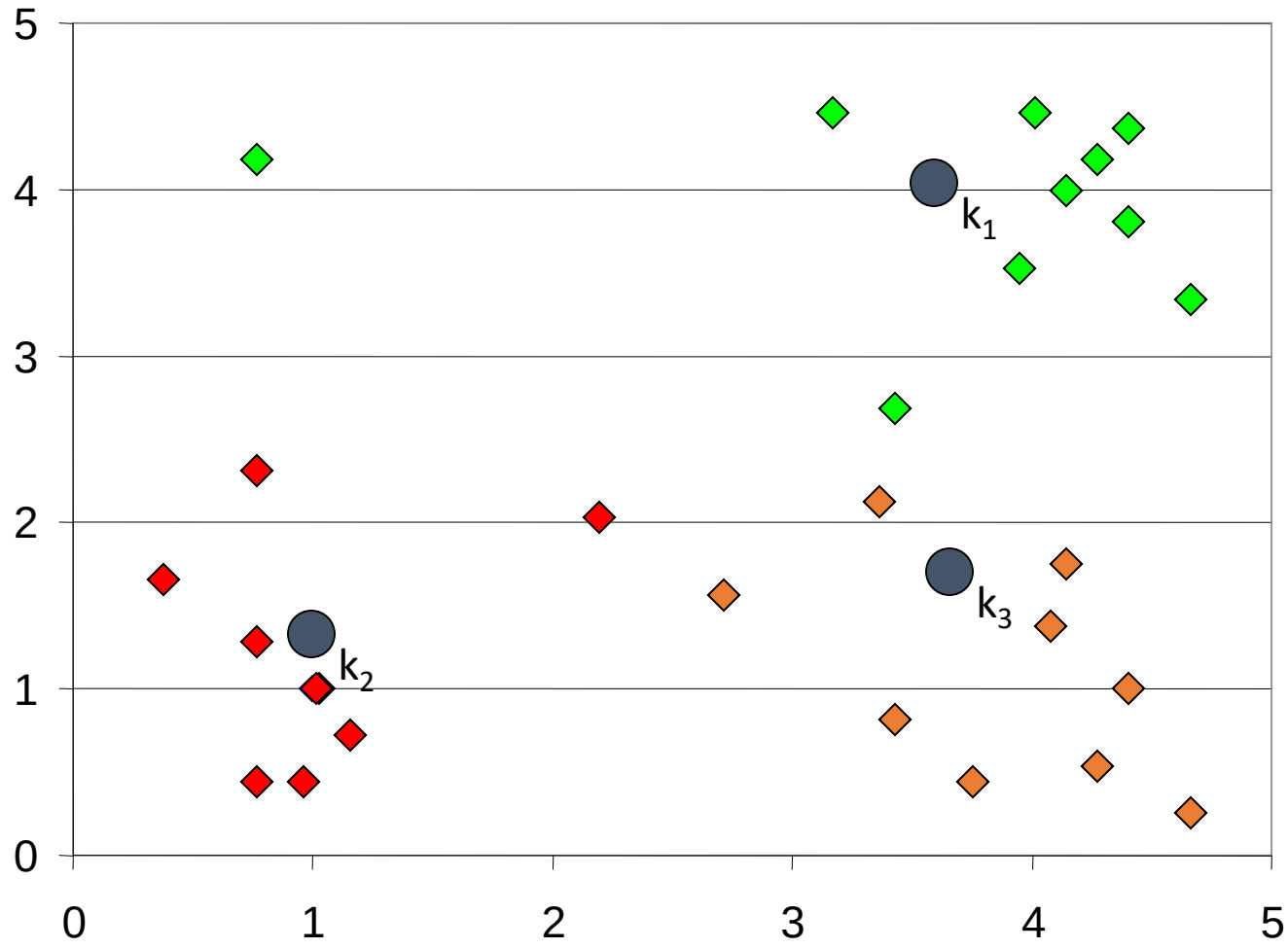
K-means Clustering: Step 1



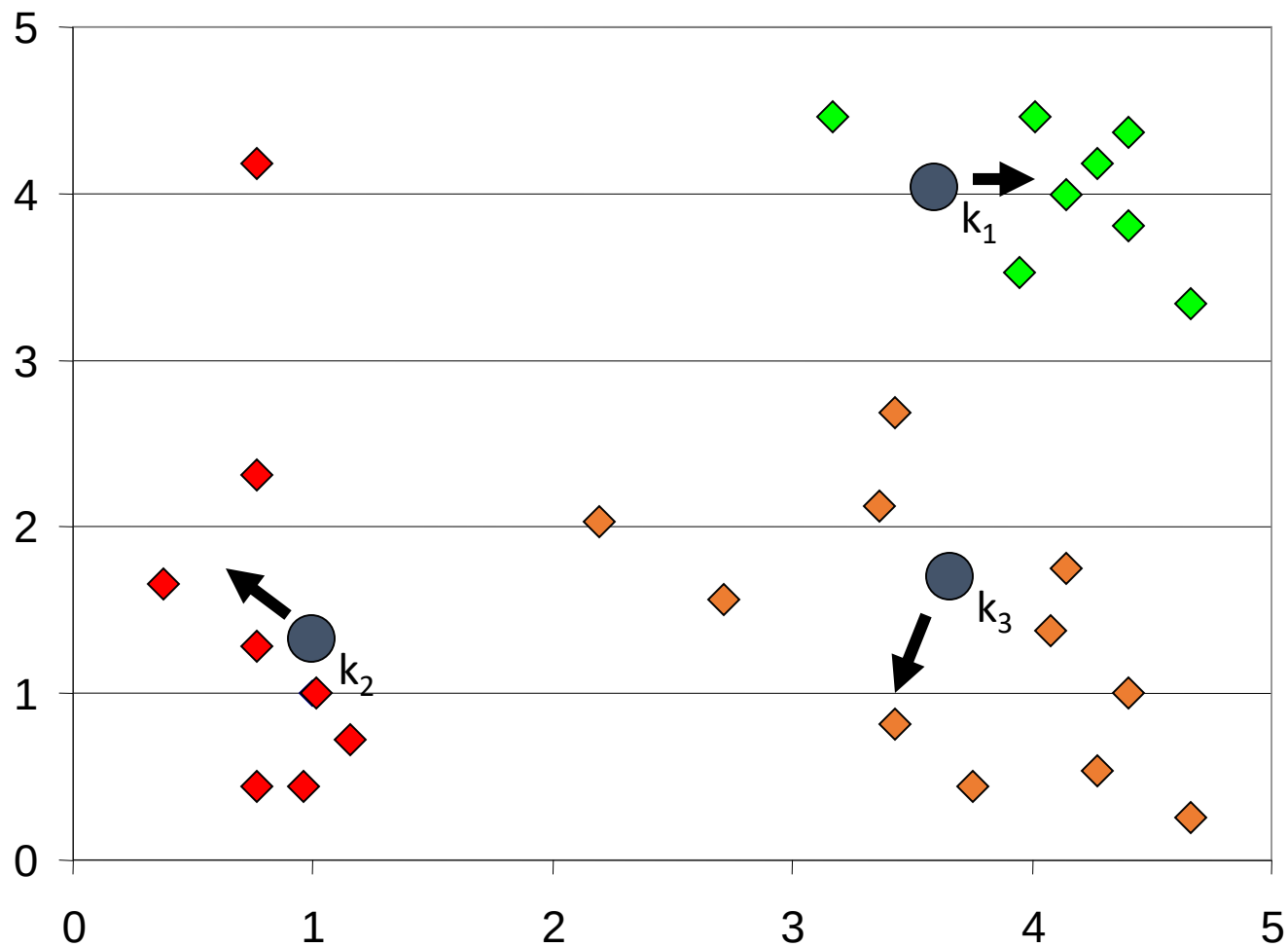
K-means Clustering: Step 2



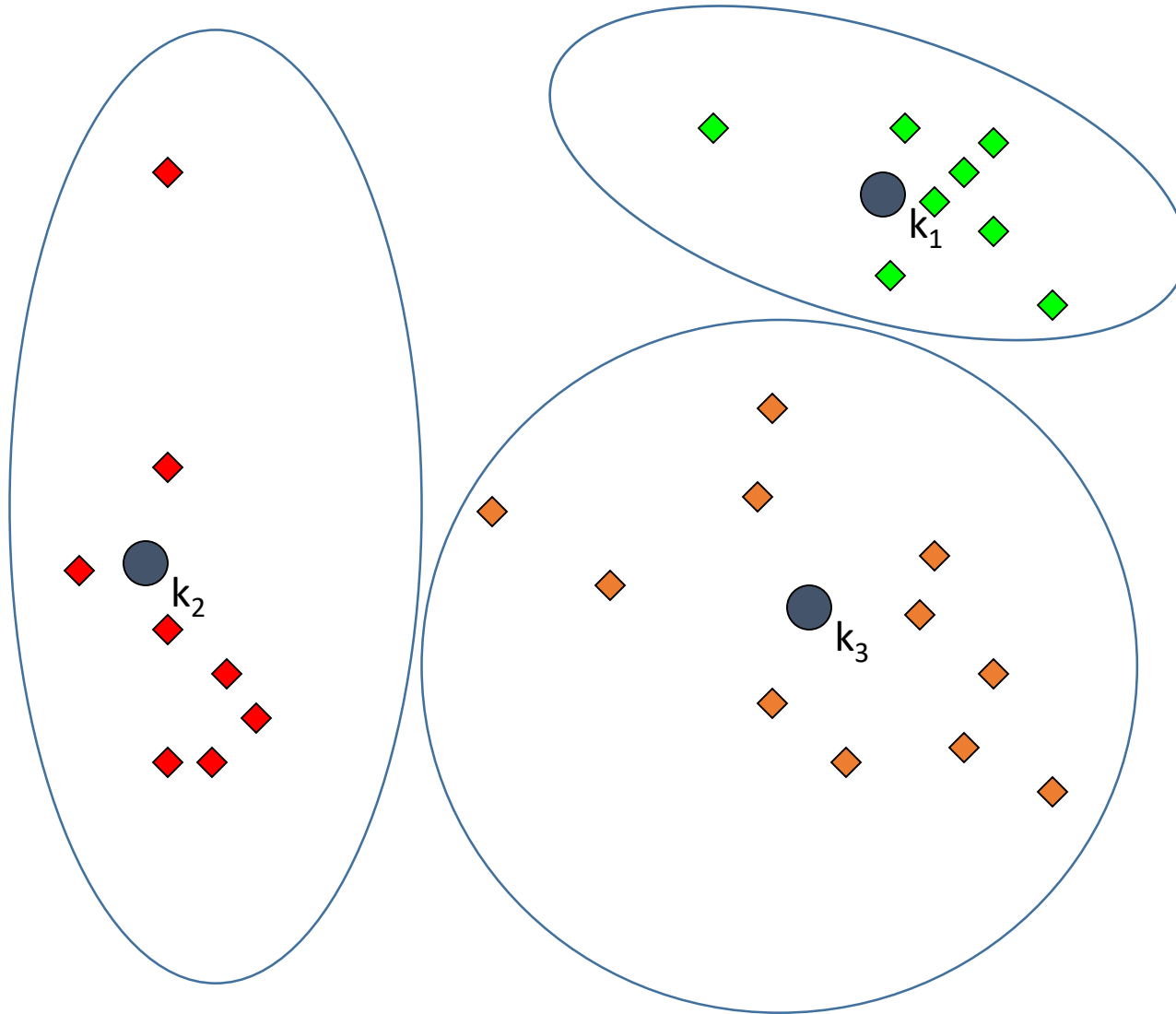
K-means Clustering: Step 3



K-means Clustering: Step 4



K-means Clustering: Step 5



Comments on k-Means algorithm (k?)

- We may not have an idea about the possible number of clusters for high dimensional data, and for data that are not scatter-plotted.
- Normally $k \ll n$ and there is heuristic to follow $k \approx \sqrt{n}$.
- There is no principled way to know what the value of k ought to be. We may try with successive value of k starting with 2 using two criterion:
- The Elbow Method
 - Calculate the Within-Cluster-Sum of Squared Errors (WSS) for different values of k, and choose the k for which WSS becomes first starts to diminish.
- The Silhouette Method
 - A high value is desirable and indicates that the point is placed in the correct cluster for different values of k.

Comments on k-Means algorithm

Distance Measurement:

- To assign a point to the closest centroid, we need a proximity measure that should quantify the notion of “closest” for the objects under clustering.
- Usually Euclidean distance (L2 norm) is the best measure when object points are defined in n-dimensional Euclidean space.
- Other measure namely cosine similarity is more appropriate when objects are of document type.
- Further, there may be other type of proximity measures that appropriate in the context of applications.
- For example, Manhattan distance (L1 norm), Jaccard measure, etc.

Comments on k-Means algorithm

Distance Measurement:

- Thus, in the context of different measures, the sum-of-squared error (i.e., objective function/convergence criteria) of a clustering can be stated as under.
- Data in Euclidean space (L2 norm):

$$SSE = \sum_{j=1}^k \sum_{X \in C_j} d(X, m_j)^2$$

- Data in Euclidean space (L1 norm):
- The Manhattan distance (L1 norm) is used as a proximity measure, where the objective is to minimize the sum-of-absolute error denoted as SAE and defined as

$$SAE = \sum_{j=1}^k \sum_{X \in C_j} |d(X, m_j)|$$

Comments on k-Means algorithm

Note:

- When SSE (L2 norm) is used as objective function and the objective is to minimize, then the cluster centroid (i.e. mean) is the mean value of the objects in the cluster.
- When the objective function is defined as SAE (L_1 norm), minimizing the objective function implies the cluster centroid as the median of the cluster.

Comments on k-Means algorithm

We know

$$SSE = \sum_{j=1}^k \sum_{X \in C_j} d(X, m_j)^2 = \sum_{j=1}^k \sum_{x \in C_j} (m_j - x)^2$$

To minimize SSE means, $\frac{\partial(SSE)}{\partial m_j} = 0$

Thus,

$$\frac{\partial}{\partial m_j} \left(\sum_{j=1}^k \sum_{x \in C_j} (m_j - x)^2 \right) = 0$$

Or,

$$\sum_{j=1}^k \sum_{x \in C_j} \frac{\partial}{\partial m_j} (m_j - x)^2 = 0$$

Comments on k-Means algorithm

Or,

$$\sum_{x \in C_j} 2(m_j - x) = 0$$

Or,

$$n_j \cdot m_j = \sum_{x \in C_j} x$$

$$m_j = \frac{1}{n_j} \sum_{x \in C_j} x$$

- Thus, the best centroid for minimizing SSE of a cluster is the mean of the objects in the cluster.
- In a Similar way, the best centroid for minimizing SAE of a cluster is the median of the objects in the cluster.

Why use K-means?

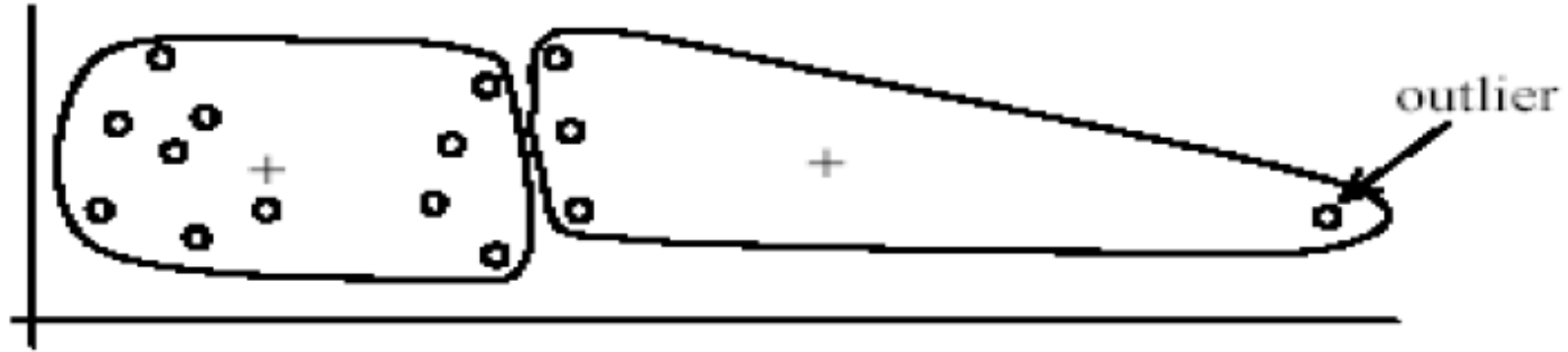
Strengths:

- Simple: easy to understand and to implement
- Efficient: Time complexity: $O(tkn)$,
 - where n is the number of data points,
 - k is the number of clusters, and
 - t is the number of iterations.
- Since both k and t are small. k-means is considered a linear algorithm.
- K-means is the most popular clustering algorithm.
- Note that: it terminates at a local optimum if SSE is used.

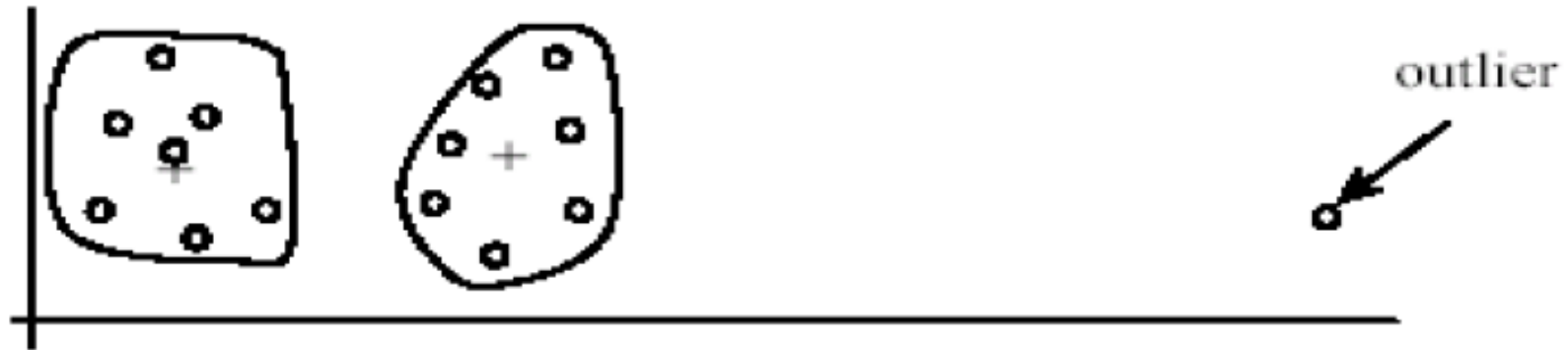
Weaknesses of K-means

- The algorithm is only applicable if the mean is defined.
- For example, k-Means does not work on categorical data because mean cannot be defined.
- The user needs to specify k.
- Unable to handle noisy data and outliers
 - Outliers are data points that are very far away from other data points.
 - Outliers could be errors in the data recording or some special data points with very different values.
- Not suitable to discover clusters with non-convex shapes

Outliers

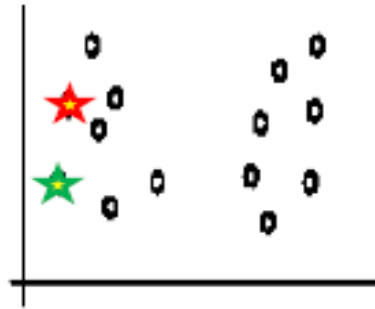


Undesirable Clusters

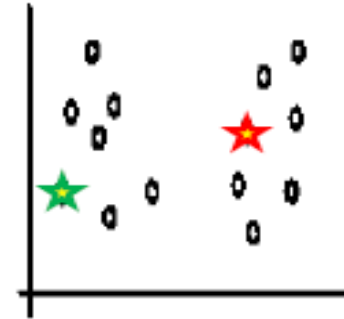


Ideal Clusters

Sensitivity to initial seeds



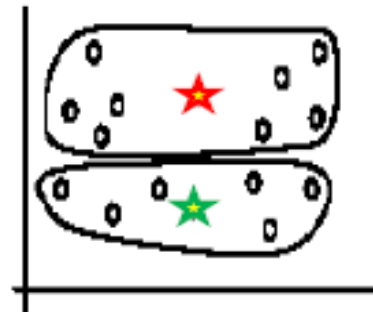
Random selection of seeds (centroids)



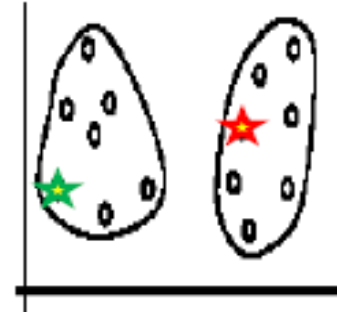
Random selection of seeds (centroids)



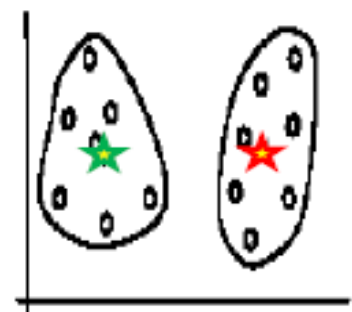
Iteration 1



Iteration 2



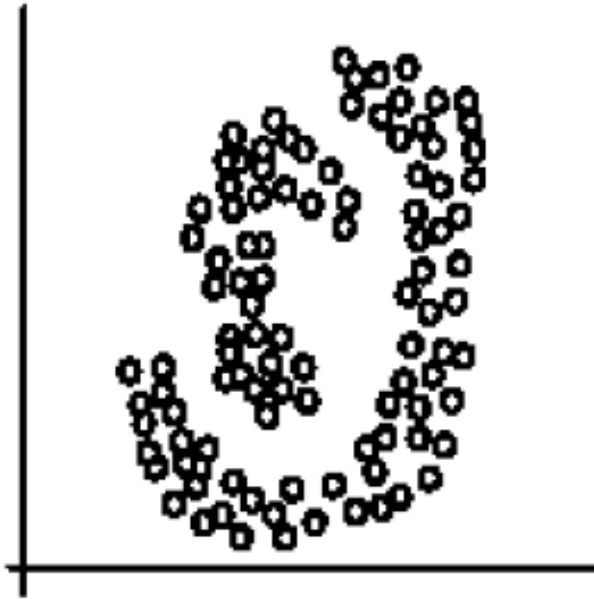
Iteration 1



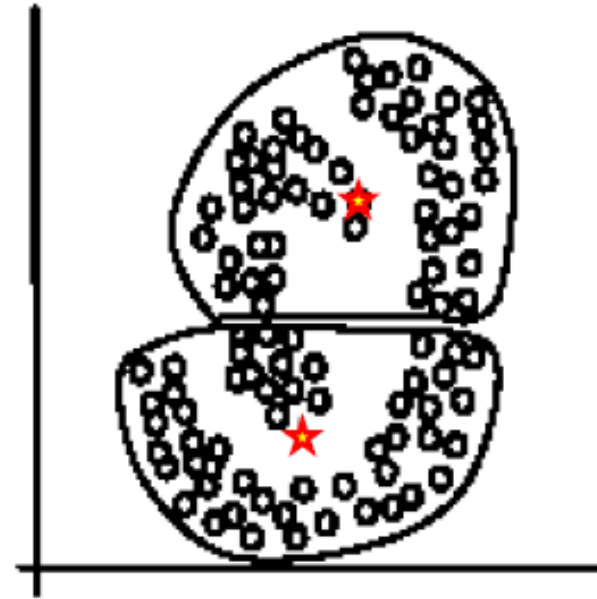
Iteration 2

Special data structures

- The k-means algorithm is not suitable for discovering clusters with non-convex shapes



Two Natural Clusters



K-means Clusters

Different variants of k-means algorithm

- There are a quite few variants of the k-Means algorithm. These can differ in the procedure of selecting the initial k means, the calculation of proximity and strategy for calculating cluster means.

Few variant of k-Means algorithm includes:

- Bisecting k-Means (addressing the issue of initial choice of cluster means).
 - M. Steinbach, G. Karypis and V. Kumar “A comparison of document clustering techniques”, *Proceedings of KDD workshop on Text mining*, 2000.
- Mean of clusters (Proposing various strategies to define means and variants of means).
 - B. zhan “Generalised k-Harmonic means – Dynamic weighting of data in unsupervised learning”, *Technical report, HP Labs*, 2000.
 - A. D. Chaturvedi, P. E. Green, J. D. Carroll, “k-Modes clustering”, *Journal of classification*, Vol. 18, PP. 35-36, 2001.
 - D. Pelleg, A. Moore, “x-Means: Extending k-Means with efficient estimation of the number of clusters”, *17th International conference on Machine Learning*, 2000.

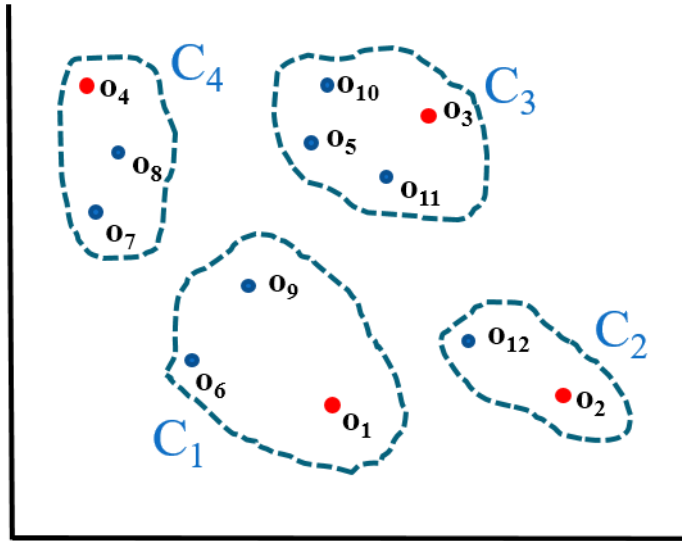
The K-Medoids Clustering Method

- Now, we shall study a variant of partitioning algorithm called k-Medoids algorithm.
 - Medoids are similar in concept to means or centroids, but medoids are always restricted to be members of the data set. Medoids are most commonly used on data when a mean or centroid cannot be defined, such as graphs.
- Motivation: We have learnt that the k-Means algorithm is sensitive to outliers because an object with an “extremely large value” may substantially distort the distribution. The effect is particularly exacerbated due to the use of the SSE (sum-of-squared error) objective function. The k-Medoids algorithm aims to diminish the effect of outliers.

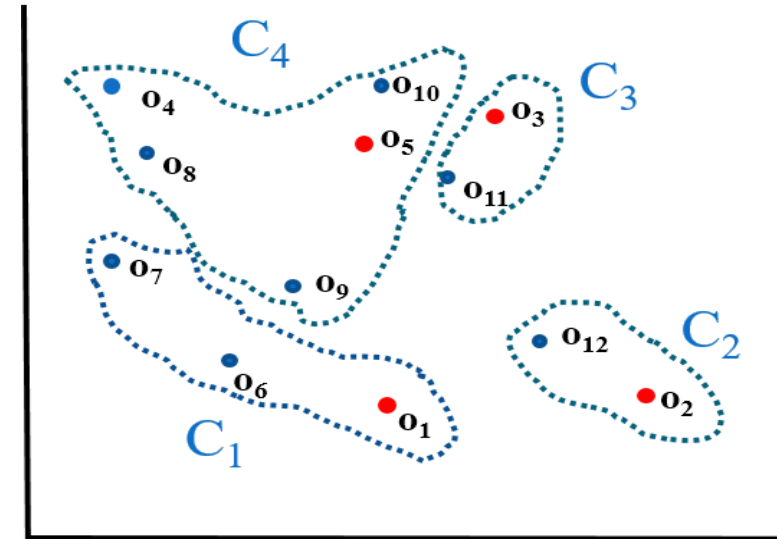
Basic concepts: K-Medoids Clustering

- The basic concepts of this algorithm is to select an object as a cluster center (one representative object per cluster) instead of taking the mean value of the objects in a cluster (as in k-Means algorithm).
- We call this cluster representative as a cluster medoid or simply medoid.
 - Initially, it selects a random set of k objects as the set of medoids.
 - Then at each step, all objects from the set of objects, which are not currently medoids are examined one by one to see if they should be medoids.
- The sum-of-absolute error (SAE) function is used as the objective function.
- The procedure terminates, if there is no any change in SAE in successive iteration (i.e. there is no change in medoid).
- This k-Medoids algorithm is also known as PAM (Partitioning around Medoids).

PAM (Partitioning around Medoids)



(a) Cluster with o_1, o_2, o_3 , and o_4 as medoids



(b) Cluster after swapping o_4 and o_5 (o_5 becomes the new medoid).

The k-Medoid method is more robust than k-Means in the presence of outliers, because a medoid is less influenced by outliers than a mean.

Thank You